

Physical Sciences

Paper 1 — Week 1 Teaching Guide

Scientific Notation · SI Base Units · SI Prefixes

Beginner to Mastery

SGL Developments | UCT Mechatronics Preparation

How to use this guide

Read each section carefully before watching any videos.

Work through every example with pen and paper — do not just read.

Complete the practice questions at the end before checking answers.

Tick every item on the Mastery Check before moving to the next section.

If anything is unclear, search the exact topic name on YouTube.

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Chapter 1 — Why This Matters for Engineering

Before you can solve a single physics problem you need to speak the language of physics. That language has two components: **scientific notation** and **SI units**. Every formula you will ever use in Physical Sciences — and later in Mechatronics Engineering — assumes your numbers are in the correct units and written in a manageable form.

Consider these real engineering numbers:

Quantity	Ordinary Number	Scientific Notation
Distance from Earth to Sun	150 000 000 000 m	1.5×10^{11} m
Diameter of a hydrogen atom	0.000 000 000 106 m	1.06×10^{-10} m
Charge of one electron	0.000 000 000 000 000 000 16 C	1.6×10^{-19} C
Speed of light	299 792 458 m/s	3.0×10^8 m/s

Scientific notation makes these numbers manageable. SI units ensure that everyone on the planet uses the same measurement system. Without both, engineering calculations become impossible to check and errors become invisible.

NOTE: Every answer in Physical Sciences must be in SI units unless the question specifically asks for a different unit.

Chapter 2 — Scientific Notation (Standard Form)

2.1 What is Scientific Notation?

Scientific notation is a way of writing any number as a value between 1 and 10, multiplied by a power of 10. It handles both very large and very small numbers cleanly and is used in every branch of science and engineering.

The format is always:

$$a \times 10^n$$

Rule	Explanation	Example
a must be ≥ 1 and < 10	Only one non-zero digit before the decimal point	3.45 correct 34.5 wrong 0.345 wrong
n is an integer	n can be positive (large number) or negative (small number)	$10^3 = 1\,000$ $10^{-3} = 0.001$
n counts the decimal move	Moving decimal LEFT gives positive n. Moving RIGHT gives negative n.	$4\,500 \rightarrow 4.5 \times 10^3$ (moved 3 left)

2.2 Converting TO Scientific Notation — Four Steps

Follow these four steps every time. Do not skip any step.

Step 1	Write the number without spaces.
Step 2	Find where the decimal point currently is. If there is no decimal point it sits at the far right of the number.
Step 3	Move the decimal point until exactly one non-zero digit sits to its left. Count how many places you moved it.
Step 4	Write the result as $a \times 10^n$ where n equals the number of places moved. n is positive if you moved LEFT. n is negative if you moved RIGHT.

2.3 Worked Examples — Converting TO Scientific Notation

Example 1: Convert 450 000 000 to scientific notation

Decimal is at the far right: 450 000 000.

Move decimal LEFT 8 places to get 4.5

n = +8 (moved left)

Answer: 4.5×10^8

Example 2: Convert 0.000 000 072 to scientific notation

Decimal is already present: 0.000000072

Move decimal RIGHT 8 places to get 7.2

$n = -8$ (moved right)

Answer: 7.2×10^{-8}

Example 3: Convert 3 200 000 000 000 to scientific notation

Move decimal LEFT 12 places: 3.2

$n = +12$

Answer: 3.2×10^{12}

Example 4: Convert 0.000 45 to scientific notation

Move decimal RIGHT 4 places: 4.5

$n = -4$

Answer: 4.5×10^{-4}

2.4 Converting FROM Scientific Notation

This is the reverse process. The exponent tells you how many places to move the decimal and in which direction.

Exponent	What to do	Example
Positive ($n > 0$)	Move decimal point RIGHT n places. Number gets LARGER.	$3.4 \times 10^5 \rightarrow$ move right 5 $\rightarrow$ 340 000
Negative ($n < 0$)	Move decimal point LEFT $ n $ places. Number gets SMALLER.	$3.4 \times 10^{-5} \rightarrow$ move left 5 $\rightarrow$ 0.000 034

Example 5: Convert 6.02×10^{23} to an ordinary number

Exponent = +23, so move decimal RIGHT 23 places

602 000 000 000 000 000 000

(This is Avogadro's number — the number of atoms in one mole)

Example 6: Convert 1.67×10^{-27} to an ordinary number

Exponent = -27, so move decimal LEFT 27 places

0.000 000 000 000 000 000 000 001 67

(This is the mass of a proton in kilograms)

WARNING: Common mistake: writing 34.5×10^7 instead of 3.45×10^8 . Always check that the number before $\times$ is between 1 and 10.

MASTERY CHECK — tick each box when you can do it confidently

- | | |
|--------------------------|---|
| ☐ | I can convert any large number to scientific notation without a calculator. |
| ☐ | I can convert any small decimal to scientific notation without a calculator. |
| ☐ | Positive exponents mean large numbers. Negative exponents mean small numbers. |
| ☐ | I can convert from scientific notation back to an ordinary number. |
| ☐ | I never write a number like 34.5×10^7 — I always adjust so the first part is between 1 and 10. |

Chapter 3 — SI Base Units

3.1 What are SI Units?

SI stands for *Système International d'Unités* — the International System of Units. It is the globally agreed measurement system used in all science and engineering. There are seven base units. Everything else — speed, force, energy, power — is derived from combinations of these seven.

3.2 The Seven SI Base Units

Quantity	Unit Name	Symbol	What it measures
Length	metre	m	Distance, displacement, wavelength
Mass	kilogram	kg	Amount of matter in an object
Time	second	s	Duration of events
Electric current	ampere	A	Flow of electric charge
Temperature	kelvin	K	Thermal energy of particles
Amount of substance	mole	mol	Number of particles (atoms, molecules)
Luminous intensity	candela	cd	Intensity of light

NOTE: The four you will use most in Physical Sciences Paper 1: metre (m), kilogram (kg), second (s), ampere (A). Know all seven, but master these four.

3.3 Why Kilograms and not Grams?

You might wonder why the base unit is the kilogram rather than the gram. This is a historical convention — when the metric system was created, the kilogram was defined as the mass of one litre of water. The important consequence for you: when you solve a formula, your mass must be in kilograms, not grams. Converting grams to kilograms is one of the most common sources of errors in exam answers.

WARNING: Never substitute mass in grams into a formula. Always convert to kilograms first: divide grams by 1 000.

3.4 Derived Units — Built from Base Units

Most quantities in physics are derived units — combinations of the base units.

Quantity	Symbol	Derived Unit	Base Unit Expression
Speed	v	m/s	$\text{m} \cdot \text{s}^{-1}$
Acceleration	a	m/s ²	$\text{m} \cdot \text{s}^{-2}$

Force	F	newton (N)	$\text{kg} \cdot \text{m} \cdot \text{s}^{-2}$
Energy/Work	E, W	joule (J)	$\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$
Power	P	watt (W)	$\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3}$
Charge	Q	coulomb (C)	$\text{A} \cdot \text{s}$
Voltage	V	volt (V)	$\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3} \cdot \text{A}^{-1}$
Resistance	R	ohm (Ω)	$\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3} \cdot \text{A}^{-2}$
Frequency	f	hertz (Hz)	s^{-1}

MASTERY CHECK — tick each box when you can do it confidently

- ☐ I can name all seven SI base units with their symbols from memory.
- ☐ I know the four base units most used in Paper 1 (m, kg, s, A).
- ☐ I always convert mass to kilograms before substituting into a formula.
- ☐ I can name at least 5 derived units and recognise their base unit expressions.

Chapter 4 — SI Prefixes

4.1 What are Prefixes?

A prefix is a word or symbol placed before a unit to indicate a multiple or fraction of that unit. Instead of writing 0.000 001 metres we write 1 micrometre (1 μm). Prefixes make measurements readable. You must know the six prefixes required by the CAPS syllabus by heart.

4.2 The Six Required Prefixes

Prefix	Symbol	Power	Multiplier	Memory anchor
giga	G	10^9	1 000 000 000	Gigabyte on your hard drive
mega	M	10^6	1 000 000	Megapixel camera
kilo	k	10^3	1 000	Kilometre on a road sign
milli	m	10^{-3}	0.001	Millimetre on a ruler
micro	μ	10^{-6}	0.000 001	Micrometre — width of a cell
nano	n	10^{-9}	0.000 000 001	Nanometre — size of atoms

NOTE: Memory trick — largest to smallest: Giga Mega Kilo [base] Milli Micro Nano. Think: 'Great Mighty Kings Make Many Miniature Nanobots'

4.3 The Prefix Number Line

Think of the prefixes as positions on a scale. Moving from nano toward giga means the numbers get larger. Each major step is a factor of 1 000 (10^3).

nano n 10^{-9}	micro μ 10^{-6}	milli m 10^{-3}	BASE unit 10^0	kilo k 10^3	mega M 10^6	giga G 10^9
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smaller values (fractions) <-- | --> larger values (multiples)

Chapter 5 — Converting Between Prefixes

5.1 The Golden Method — Always Go Through the Base Unit

When converting between two prefixed units the safest method is always to convert to the base unit first, then to the target unit. This prevents errors and works for every conversion without exception.

The Two-Step Method

Step 1: Convert from the given prefix to the base unit.

Multiply by the prefix value (the power of 10).

Step 2: Convert from the base unit to the target prefix.

Divide by the target prefix value.

Alternatively: express everything in scientific notation and compare powers of 10.

5.2 Converting TO the Base Unit

To remove a prefix and get the base unit value, multiply by the prefix's power of 10.

From prefix	Operation	Example
giga (G)	$\times 10^9$	3.2 GB $\rightarrow 3.2 \times 10^9$ bytes
mega (M)	$\times 10^6$	8.5 MHz $\rightarrow 8\,500\,000$ Hz
kilo (k)	$\times 10^3$	5 km $\rightarrow 5\,000$ m
milli (m)	$\times 10^{-3}$	350 mm $\rightarrow 350 \times 10^{-3}$ m = 0.350 m
micro (μ)	$\times 10^{-6}$	62 μ m $\rightarrow 62 \times 10^{-6}$ m = 6.2×10^{-5} m
nano (n)	$\times 10^{-9}$	450 nm $\rightarrow 450 \times 10^{-9}$ m = 4.5×10^{-7} m

5.3 Converting FROM the Base Unit to a Prefix

To express a base unit value in a given prefix, divide by the prefix's power of 10.

Target prefix	Operation	Example
giga (G)	$\div 10^9$	3 200 000 000 Hz = 3.2 GHz
mega (M)	$\div 10^6$	5 000 000 W = 5 MW
kilo (k)	$\div 10^3$	7 500 m = 7.5 km
milli (m)	$\div 10^{-3} = \times 10^3$	0.025 m = 25 mm
micro (μ)	$\div 10^{-6} = \times 10^6$	0.000 062 m = 62 μ m

nano (n)	$\div 10^{-9} = \times 10^9$	$4.5 \times 10^{-7} \text{ m} = 450 \text{ nm}$
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WARNING: Watch the direction: converting TO a smaller unit (e.g. metres $\rightarrow$ millimetres) makes the number LARGER. Converting TO a larger unit (e.g. metres $\rightarrow$ kilometres) makes the number SMALLER. If your answer goes the wrong way, re-check.

MASTERY CHECK — tick each box when you can do it confidently

- ☐ I know all six required prefixes with their symbol and power of 10.
- ☐ I can convert any prefixed unit to the base unit.
- ☐ I can convert any base unit value to any prefixed unit.
- ☐ I understand that converting to a smaller unit gives a larger number.
- ☐ I use the two-step method (prefix $\rightarrow$ base $\rightarrow$ target prefix) for tricky conversions.

Chapter 6 — Worked Examples Step by Step

Cover the solution and try each problem yourself first. Only look at the solution when you are stuck or want to verify.

Example 1 — km to m

Question: Convert 5 km to metres.

Solution:

- Identify the prefix: kilo (k) = 10^3
 - To remove the prefix, multiply by 10^3
 - $5 \text{ km} = 5 \times 10^3 \text{ m}$
 - ✓ Answer: 5 000 m
-

Example 2 — mm to m

Question: Convert 350 mm to metres.

Solution:

- Identify the prefix: milli (m) = 10^{-3}
 - To remove the prefix, multiply by 10^{-3}
 - $350 \text{ mm} = 350 \times 10^{-3} \text{ m}$
 - ✓ Answer: 0.350 m
-

Example 3 — m to mm

Question: Convert 0.0025 m to millimetres.

Solution:

- Target prefix: milli = 10^{-3}
 - Divide by 10^{-3} (same as multiplying by 10^3)
 - $0.0025 \div 10^{-3} = 0.0025 \times 10^3$
 - ✓ Answer: 2.5 mm
-

Example 4 — Hz to MHz

Question: Convert 8 500 000 Hz to megahertz.

Solution:

→ Target prefix: mega = 10^6

→ Divide by 10^6

→ $8\,500\,000 \div 10^6 = 8.5$

✓ Answer: 8.5 MHz

Example 5 — GB to bytes

Question: Convert 3.2 GB to bytes.

Solution:

→ Prefix: giga = 10^9

→ Multiply by 10^9

→ 3.2×10^9

✓ Answer: 3 200 000 000 bytes

Example 6 — nm to m

Question: Convert 450 nm to metres.

Solution:

→ Prefix: nano = 10^{-9}

→ Multiply by 10^{-9}

→ $450 \times 10^{-9} = 4.50 \times 10^{-7}$

✓ Answer: 4.5×10^{-7} m

Example 7 — m to μm

Question: Convert 0.000 062 m to micrometres.

Solution:

→ Target prefix: micro = 10^{-6}

→ Divide by 10^{-6} (multiply by 10^6)

→ $0.000\,062 \times 10^6 = 62$

✓ Answer: 62 μm

Example 8 — mg to kg (two-step)

Question: Convert 7 500 mg to kilograms.

Solution:

→ Step 1: mg → g. milli = 10^{-3} , so $7\,500\text{ mg} = 7\,500 \times 10^{-3}\text{ g} = 7.5\text{ g}$

→ Step 2: g → kg. kilo = 10^3 , so divide by 10^3 : $7.5 \div 10^3 = 0.0075$

→ Answer: 0.0075 kg

✓ Check: $0.0075\text{ kg} \times 10^3\text{ g/kg} \times 10^3\text{ mg/g} = 7\,500\text{ mg}$ correct

Example 9 — scientific notation prefix conversion

Question: Convert 2.4×10^{-3} km to millimetres.

Solution:

→ Step 1: km → m. kilo = 10^3 , multiply by 10^3

→ $2.4 \times 10^{-3} \times 10^3\text{ m} = 2.4 \times 10^0\text{ m} = 2.4\text{ m}$

→ Step 2: m → mm. milli = 10^{-3} , divide by 10^{-3}

→ $2.4 \div 10^{-3} = 2.4 \times 10^3 = 2\,400$

✓ Answer: 2 400 mm

Example 10 — real world

Question: A bacterium is $2\text{ }\mu\text{m}$ long. Express this in metres using scientific notation.

Solution:

→ Prefix: micro (μ) = 10^{-6}

→ Multiply by 10^{-6}

→ 2×10^{-6}

→ This is already in correct scientific notation (2 is between 1 and 10).

✓ Answer: $2 \times 10^{-6}\text{ m}$

Chapter 7 — Practice Questions (45 Questions)

Complete all questions with pen and paper. Do not use a calculator for scientific notation conversions. Show all working. Check answers in Chapter 8 only after attempting each question independently.

WARNING: Do not skip questions. Difficulty increases progressively. If you can answer question 45 correctly, you have mastered this topic.

Section A — Scientific Notation: Convert to Standard Form (10 questions)

A1.

Convert 450 000 000 to scientific notation.

A2.

Convert 0.000 000 072 to scientific notation.

A3.

Convert 3 200 000 000 000 to scientific notation.

A4.

Convert 0.000 45 to scientific notation.

A5.

Convert 93 000 000 to scientific notation.

A6.

Convert 0.000 000 000 000 000 16 to scientific notation.

A7.

Convert 12 700 000 to scientific notation.

A8.

Convert 0.000 308 to scientific notation.

A9.

Convert 56 000 000 000 to scientific notation.

A10.

Convert 0.000 000 000 911 to scientific notation.

Section B — Scientific Notation: Convert from Standard Form (10 questions)

B1.

Convert 6.02×10^{23} to an ordinary number.

B2.

Convert 3.4×10^{-5} to an ordinary number.

B3.

Convert 1.67×10^{-27} to an ordinary number.

B4.

Convert 9.1×10^6 to an ordinary number.

B5.

Convert 2.998×10^8 to an ordinary number.

B6.

Convert 4.8×10^{-3} to an ordinary number.

B7.

Convert 7.35×10^{22} to an ordinary number.

B8.

Convert 1.0×10^{-10} to an ordinary number.

B9.

Convert 3.84×10^8 to an ordinary number.

B10.

Convert 5.0×10^{-7} to an ordinary number.

Section C — Scientific Notation: Mixed Operations (5 questions)

C1.

A radio wave has a frequency of 98 000 000 Hz. Express this in scientific notation.

C2.

A virus is 2.0×10^{-7} m in diameter. Write this as an ordinary decimal number.

C3.

The distance to Proxima Centauri is 40 200 000 000 000 km. Express in scientific notation.

C4.

Light travels 9.461×10^{12} km in one year. Write this as an ordinary number.

C5.

An electron has a mass of 0.000 000 000 000 000 000 000 000 911 kg. Express in scientific notation.

Section D — SI Unit Conversions (10 questions)

D1.

Convert 5 km to metres.

D2.

Convert 350 mm to metres.

D3.

Convert 0.0025 m to millimetres.

D4.

Convert 8 500 000 Hz to megahertz (MHz).

D5.

Convert 3.2 GB to bytes.

D6.

Convert 450 nm to metres. Give your answer in scientific notation.

D7.

Convert 0.000 062 m to micrometres (μm).

D8.

Convert 7 500 mg to kilograms.

D9.

Convert 2.4×10^{-3} km to millimetres.

D10.

A bacterium is 2 μm long. Express this in metres using scientific notation.

Section E — Advanced Prefix Conversions (10 questions)

E1.

Convert 0.045 km to centimetres.

E2.

A signal has a wavelength of 5×10^{-7} m. Express this in nanometres.

E3.

A capacitor stores 4 700 μF (microfarads). Convert this to farads.

E4.

Convert 2.5 GHz to Hz, then write your answer in scientific notation.

E5.

A nano-particle has a diameter of 80 nm. Express this in metres and in micrometres.

E6.

Convert 0.5 MJ (megajoules) to joules.

E7.

A current of 3.5 mA flows through a circuit. Convert this to amperes.

E8.

Convert 1 200 000 000 nm to metres, then to km.

E9.

A protein is 5×10^{-9} m wide. Express this in nanometres.

E10.

Convert 6.5×10^7 μg to kilograms.

Chapter 8 — Full Answer Key

Section A — Convert to Scientific Notation

A1	4.5×10^8
A2	7.2×10^{-8}
A3	3.2×10^{12}
A4	4.5×10^{-4}
A5	9.3×10^7
A6	1.6×10^{-19}
A7	1.27×10^7
A8	3.08×10^{-4}
A9	5.6×10^{10}
A10	9.11×10^{-31}

Section B — Convert from Scientific Notation

B1	602 000 000 000 000 000 000
B2	0.000 034
B3	0.000 000 000 000 000 000 000 001 67
B4	9 100 000
B5	299 800 000
B6	0.0048
B7	73 500 000 000 000 000 000 000
B8	0.000 000 000 1
B9	384 000 000
B10	0.000 000 5

Section C — Mixed Scientific Notation

C1	9.8×10^7 Hz
C2	0.000 000 2 m
C3	4.02×10^{13} km
C4	9 461 000 000 000 km
C5	9.11×10^{-31} kg

Section D — SI Unit Conversions

D1	5 000 m
D2	0.350 m
D3	2.5 mm
D4	8.5 MHz
D5	3 200 000 000 bytes = 3.2×10^9 bytes
D6	4.5×10^{-7} m
D7	62 μm
D8	0.0075 kg
D9	2 400 mm
D10	2×10^{-6} m

Section E — Advanced Prefix Conversions

E1	$0.045 \text{ km} \times 10^3 = 45 \text{ m} \times 10^2 = 4\,500 \text{ cm}$
E2	$5 \times 10^{-7} \text{ m} \div 10^{-9} = 500 \text{ nm}$
E3	$4\,700 \times 10^{-6} \text{ F} = 4.7 \times 10^{-3} \text{ F} = 0.0047 \text{ F}$
E4	$2.5 \times 10^9 \text{ Hz} = 2\,500\,000\,000 \text{ Hz}$
E5	$80 \times 10^{-9} \text{ m} = 8.0 \times 10^{-8} \text{ m}$; also = $0.08 \mu\text{m}$
E6	$0.5 \times 10^6 \text{ J} = 500\,000 \text{ J}$
E7	$3.5 \times 10^{-3} \text{ A} = 0.0035 \text{ A}$
E8	1.2 m; $0.0012 \text{ km} = 1.2 \times 10^{-3} \text{ km}$
E9	$5 \times 10^{-9} \text{ m} \div 10^{-9} = 5 \text{ nm}$
E10	$6.5 \times 10^7 \times 10^{-6} \text{ g} = 65 \text{ g} = 0.065 \text{ kg}$

Chapter 9 — Quick Reference Card

Photograph this page and keep it visible while you study.

Scientific Notation Rules

Rule	Detail
Format	$a \times 10^n$ where $1 \leq a < 10$
Large numbers (> 1)	Decimal moves LEFT → n is POSITIVE
Small numbers (< 1)	Decimal moves RIGHT → n is NEGATIVE
Positive exponent	Move decimal RIGHT to get the ordinary number
Negative exponent	Move decimal LEFT to get the ordinary number

SI Prefixes

giga	G	10^9	1 000 000 000	3.2 GB
mega	M	10^6	1 000 000	8.5 MHz
kilo	k	10^3	1 000	5 km
BASE	—	10^0	1	metre, kg, second
milli	m	10^{-3}	0.001	350 mm
micro	μ	10^{-6}	0.000 001	2 μm
nano	n	10^{-9}	0.000 000 001	450 nm

Seven SI Base Units

Quantity	Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

NOTE: Conversion shortcut: Prefix → Base: MULTIPLY by prefix value. Base → Prefix: DIVIDE by prefix value.